TSM 099 Polynomials Test

Practice 2 | Version 202610122020.20

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right.

QUESTIONS	ANSWERS
1. Factor completely. $(a)\quad x^2+2x-15$ $(b)\quad 4x^2-9$	1. (a) 10 pts (b) 10 pts
2. Factor completely over the integers, or write prime. $(a)\quad 3x^2+2x+1$ $(b)\quad 3x^2+x-2$	2. (a) 10 pts (b) 10 pts
3. Factor out the greatest common factor. $6x^4-3x^3+12x^2$	3. Factored form 20 pts
4. Multiply and simplify. $(3x-3)(4x+2)$	4. Polynomial 20 pts
5. Factor using the ac-method and grouping. Show ac, the split middle term, and the grouping. $12x^2+25x+12$	5. ac; pair 4 pts Split middle term 4 pts Grouping 6 pts Factored form 6 pts

Instructor answer key: TSM 099 Polynomials Test

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. (a)

$(x-3)(x+5)$

10 pts

(b)

$(2x-3)(2x+3)$

10 pts

2. (a)

Prime

10 pts

(b)

$(x+1)(3x-2)$

10 pts

3. Factored form

$3x^2(2x^2-x+4)$

20 pts

4. Polynomial

$12x^2-6x-6$

20 pts

5. ac; pair

$ac=144;\ 9,\ 16$

4 pts

Split middle term

$12x^2+9x+16x+12$

4 pts

Grouping

$3x(4x+3)+4(4x+3)$

6 pts

Factored form

$(3x+4)(4x+3)$

6 pts